

WHETHER ONE STEP IS ENOUGH IS NOT DETERMINED BY THE DATA: COUPLING, LOSS FORM, AND AN IMPOSSIBILITY THEOREM

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ABSTRACT

Few-step generative policies are reported both to match iterative samplers and to collapse, and the literature attributes the difference to data, architecture, or sampler. We show that “is one step enough for this task” has no answer at the level of a dataset. We study the *realised conditional-spread ratio* ρ , the fraction of conditional target variance a one-step map preserves, and show that ρ of the L^2 -optimal one-step map is a property of the *training target*—the source–target coupling together with the form of the loss—not of the data. We construct an explicit one-parameter family of training targets whose induced (c, x_1) law is the same for every α , but for which the optimal ρ^* sweeps all of $[0, 1]$ (analytically $\rho^*(\alpha) = \alpha^2$; measured 0.05006 to 0.9766). Hence the minimax risk of *any* statistic of the data—even one given the population law exactly—equals 0.5, the risk of the constant predictor: the data are worth exactly zero here, and no sample size improves this. We confirm that seven data-side statistics, two of them ours, show no detectable dependence on α while ρ^* varies by 1.0. The positive counterpart: ρ is computable from the paired samples already in a training minibatch, before any one-step model exists. The loss channel turns on the *scope* of the matching: a pointwise L^2 loss collapses ($\rho = 9.815 \times 10^{-5}$); a Chamfer-style loss matching each target against the *whole batch* recovers the *support* but flattens the conditional weights (CV 0.12 versus 0.398); matching *within* each condition restores the spread, and a balanced within-condition assignment is closest on conditional sliced W_2 ($1.635 \times$ lower). All experiments are CPU-only on closed-form or fully controlled constructions.

1 INTRODUCTION

Generative policies—diffusion and flow-matching action heads—are a default way to represent multimodal behaviour in continuous control and vision-language-action models (Chi et al., 2023; Wang et al., 2022). Their cost is iterative sampling, so a large literature compresses sampling to one or a few network evaluations: consistency models (Song et al., 2023), shortcut models (Frans et al., 2024), mean flows (Geng et al., 2025), rectified-flow reflow (Liu et al., 2022), distillation (Salimans & Ho, 2022; Yin et al., 2023), and in RL one-step flow policies (Nguyen et al., 2026; Li et al., 2026a; Zhan et al., 2026).

The reported picture is inconsistent in an instructive way: one-step policies are said to match 100-step diffusion on 56 manipulation tasks (Li et al., 2026a) and to reach 92.8 on D4RL MuJoCo (Nguyen et al., 2026), yet the same families degrade when the conditioning is weakened or the action horizon lengthened (Li et al., 2026a; Chen et al., 2026). The most explicit diagnosis frames one-step success as a property of the condition–target *structure* (Chen et al., 2026), and states as an open problem the task of *quantifying* when it holds.

This paper argues the open problem is mis-posed as stated. The quantity that decides one-step adequacy is not a property of the data distribution $p(c, x_1)$ at all: it is a property of the *training target*. We make this precise with three results and one map.

What we mean by one-step adequacy. For a one-step map f we define the realised conditional-spread ratio

$$\rho(f) = \frac{\mathbb{E}_c[\text{tr Var}_{x_0}(f(x_0, c) | c)]}{\mathbb{E}_c[\text{tr Var}(x_1 | c)]} \in [0, 1], \quad (1)$$

the fraction of the conditional target variance that survives. $\rho \approx 0$ is collapse onto a per-condition point mass; $\rho \approx 1$ is a full conditional law. Unlike a likelihood or a sample-quality score, ρ separates the two failure modes that practitioners care about—blurring versus mode dropping—in a single number that requires no trained generative model to define.

Contributions.

- C1 ρ is a property of the training target, not of the data.** The L^2 -optimal one-step map is $f^*(x_0, c) = \mathbb{E}_\pi[x_1 | x_0, c]$ (Thm. 1); $\rho(f^*)$ equals 0 under an independent coupling and 1 under a deterministic one (Cor. 1, 2).
- C2 An impossibility theorem, in minimax form.** There is a one-parameter family of training targets with an *identical* (c, x_1) law for which $\rho^*(\alpha) = \alpha^2$ sweeps $[0, 1]$ (Thm. 2). Hence $\inf_{\mathcal{A}} \sup_{\alpha} \mathbb{E}|\mathcal{A} - \rho^*(\alpha)| = 0.5$ over *all* data-only statistics, attained by the constant $1/2$ (Thm. 3). More data does not help.
- C3 The positive counterpart.** ρ is estimable from the paired samples already in a training minibatch, before the one-step model is trained; the estimator obeys an analytic $1/k$ bias law and tracks α^2 to MAD 0.01405.
- C4 Two escape channels; for the loss channel, scope decides.** One-step failure can be removed either by changing the coupling (Channel 1: OT coupling, reflow, distillation) or by changing the loss form (Channel 2). Channel 2 has a subtle structure that the literature does not state: for a collection-level loss, what decides whether the conditional *spread* is recovered is first the *scope* of the matching, then balancedness. A pooled Chamfer recovers the support but flattens the weights; matching within each condition restores the spread; a balanced within-condition assignment is closest on conditional sliced W_2 (Thm. 4, Sec. 4).
- C5 A recorded negative result about our own earlier proposal.** The shape statistic $\hat{\Lambda}$ we previously advanced as a modality detector sorts a single-mode banana above an eight-mode ring (0.2257 versus 0.09003); it is demoted to a descriptive index (Sec. 6).

Scope and honesty statement. Every experiment is CPU-only, on synthetic families with closed-form or exactly controlled ground truth. An earlier iteration trained on conditional MNIST; those runs are not reported (the field was under-fitted at our CPU budget) and nothing here depends on them. Claims about what *cannot* be predicted are exact theorems; claims about what *does* happen under a given training target are controlled measurements with pre-registered criteria. Appendix E gives the full statement.

2 SETUP, AND WHAT ρ DEPENDS ON

Let $c \in \mathcal{C}$ be a conditioning variable (observation, state, goal, embedding) and $x_1 \in \mathbb{R}^d$ a target (action, action chunk, image), with joint law $p(c, x_1)$ observed only through i.i.d. samples. A one-step generative policy is a map $f : \mathbb{R}^d \times \mathcal{C} \rightarrow \mathbb{R}^d$ applied to source noise $x_0 \sim \mathcal{N}(0, I_d)$; a flow or diffusion policy instead integrates a velocity field learned from a *coupling* π between x_0 and x_1 . Throughout, π is the joint law of (x_0, x_1) given c , with both marginals fixed: $x_0 \sim \mathcal{N}(0, I_d)$ and $x_1 \sim p(\cdot | c)$. The two couplings that matter in practice are the *independent* coupling $x_0 \perp x_1 | c$ used by vanilla conditional flow matching (Lipman et al., 2022; Tong et al., 2023), and a *deterministic* coupling $x_1 = T(x_0, c)$, which is what reflow (Liu et al., 2022) and consistency distillation produce.

Theorem 1 (Optimal one-step map, and what ρ is). *Fix a coupling π . The minimiser of the pointwise L^2 risk $\mathbb{E}_\pi \|f(x_0, c) - x_1\|^2$ over all measurable f is $f^*(x_0, c) = \mathbb{E}_\pi[x_1 | x_0, c]$, and*

$$\rho(f^*) = \frac{\mathbb{E}_c[\text{tr Var}(\mathbb{E}[x_1 | x_0, c] | c)]}{\mathbb{E}_c[\text{tr Var}(x_1 | c)]} \in [0, 1],$$

with the bound following from the law of total variance, $\text{Var}(x_1 \mid c) = \mathbb{E}[\text{Var}(x_1 \mid x_0, c) \mid c] + \text{Var}(\mathbb{E}[x_1 \mid x_0, c] \mid c)$.

Proof sketch. The conditional mean is the L^2 projection; the law of total variance gives the decomposition and hence $\rho \leq 1$; $\rho \geq 0$ is trivial. Appendix A. $\square$

Theorem 1 is a one-line consequence of the tower property, and we do not claim it as new mathematics. Its content for our purpose is the *dependence structure it exposes*: $\rho(f^*)$ is a functional of π , and only of π and of $p(\cdot \mid c)$ through the denominator. Two immediate corollaries are the endpoints of the whole story.

Corollary 1 (Independent coupling collapses). *If $x_0 \perp\!\!\!\perp x_1 \mid c$ then $\mathbb{E}[x_1 \mid x_0, c] = \mathbb{E}[x_1 \mid c] = m(c)$ is constant in x_0 , so $\rho(f^*) = 0$: the optimal one-step map is a per-condition point mass.*

Corollary 2 (Deterministic coupling does not). *If $x_1 = T(x_0, c)$ with $T(\cdot, c) \# \mathcal{N}(0, I_d) = p(\cdot \mid c)$, then $f^* = T$ and $\rho(f^*) = 1$.*

Corollary 1 in closed form. We verify the endpoint identity numerically on the exact velocity field of a Gaussian-mixture conditional, where $v^*(x, 0, c) = \mathbb{E}[x_1 \mid c] - x$ can be integrated in closed form: the unit Euler endpoint matches $m(c)$ to 8.882×10^{-16} and realises conditional variance 3.592×10^{-32} , while the 8-mode ring is covered 0 times at $N = 1$ and 8/8 from $N = 2$ on. The multi-step ratio rises monotonically,

N	1	2	4	8	16	32	64	128	256
ρ_N	3.936×10^{-33}	0.5907	0.8573	0.9233	0.9612	0.9815	0.992	0.9974	1

Under a conditional (multi-condition) version of the same construction, changing *only* the coupling moves one-step ρ from 0.02722 to 0.8776 at $k = 30$ —a factor of 32.24—while the data statistic $\hat{D}$ takes the value 0.9116 in *both* arms, to a measured difference of 6.661×10^{-16} . That invariance is exact rather than a near-miss measurement: $\hat{D}$ is a functional of the (c, x_1) pairs alone, and switching the coupling only reorders those pairs within each condition.

Two channels, stated once. Theorem 1 shows collapse is not a fact about one-step evaluation per se. It is the conjunction of two design choices:

$$\text{collapse} \iff \underbrace{\text{independent coupling}}_{\text{Channel 1}} \wedge \underbrace{\text{pointwise } L^2 \text{ loss}}_{\text{Channel 2}}. \quad (2)$$

Removing *either* conjunct removes the collapse. Channel 1 is the road the literature has taken (OT coupling, reflow, distillation). Channel 2 is not, and Sec. 4 shows it has a finer structure than has been stated.

3 THE DATA CANNOT DECIDE

The two channels in (2) are properties of how you train. A natural hope is that they are nevertheless *predictable* from the data: that some statistic of a dataset of (c, x_1) pairs tells you in advance whether one step will suffice. This section shows the hope is not merely hard but void, and quantifies exactly how void it is.

3.1 A FAMILY WITH FIXED DATA AND SWEEPING ρ

Fix $p(\cdot \mid c)$ and let $T(\cdot, c)$ be any transport map pushing $\mathcal{N}(0, I_d)$ to $p(\cdot \mid c)$ —concretely, the flow map of the exact velocity field, i.e. what reflow would produce. For $\alpha \in [0, 1]$ define the α -interpolated coupling

$$x_1 = B \cdot T(x_0, c) + (1 - B) \cdot X, \quad B \sim \text{Bernoulli}(\alpha), \quad X \sim p(\cdot \mid c), \quad X \perp\!\!\!\perp x_0 \mid c. \quad (3)$$

Theorem 2 (Fixed data, sweeping ρ). *For the coupling (3): (i) both branches have marginal $p(\cdot \mid c)$, so the law of (c, x_1) does not depend on α ; (ii) $f_\alpha^*(x_0, c) = \alpha T(x_0, c) + (1 - \alpha)m(c)$; and (iii) $\rho^*(\alpha) = \alpha^2$.*

Proof. (i) $T(x_0, c) \sim p(\cdot | c)$ because T is a transport map, and $X \sim p(\cdot | c)$ by construction; a mixture of two identical laws is that law. (ii) $\mathbb{E}[x_1 | x_0, c] = \alpha \mathbb{E}[T(x_0, c) | x_0, c] + (1 - \alpha) \mathbb{E}[X | c]$. (iii) Given c , the conditional variance of $\alpha T(x_0, c) + (1 - \alpha)m(c)$ over x_0 is $\alpha^2 \text{Var}(x_1 | c)$, since the second term is constant and $\text{Var}(T(x_0, c) | c) = \text{Var}(x_1 | c)$. $\square$

The family is not exotic: $\alpha = 0$ is the standard independent coupling of vanilla conditional flow matching, $\alpha = 1$ is the deterministic coupling that reflow produces, and intermediate α is a mixture of the two. What varies is a design choice; what does not vary is the law of a single observed pair $(c_i, x_{1,i})$.

3.2 MINIMAX FORM: THE DATA ARE WORTH EXACTLY ZERO

Theorem 3 (Impossibility, minimax form). *Let $\mathcal{A}$ be any estimator of ρ^* measurable with respect to n i.i.d. draws from $p(c, x_1)$ —randomised, and allowed to know the population law $p(c, x_1)$ exactly. Then for the family of Theorem 2,*

$$\inf_{\mathcal{A}} \sup_{\alpha \in [0,1]} \mathbb{E}_{\alpha} |\mathcal{A} - \rho^*(\alpha)| = \frac{1}{2} = \inf_{a \in \mathbb{R}} \sup_{\alpha \in [0,1]} |a - \alpha^2|,$$

and the infimum on the left is attained by the constant $\mathcal{A} \equiv 1/2$.

Proof. By Theorem 2(i) the law of the sample is the same for every α , so the law of $\mathcal{A}$ —and in particular $\bar{a} := \mathbb{E}[\mathcal{A}]$ —does not depend on α . By Jensen, $\mathbb{E}_{\alpha} |\mathcal{A} - \alpha^2| \geq |\bar{a} - \alpha^2|$ for each α , hence $\sup_{\alpha} \mathbb{E}_{\alpha} |\mathcal{A} - \alpha^2| \geq \sup_{\alpha} |\bar{a} - \alpha^2| \geq \max\{|\bar{a}|, |\bar{a} - 1|\} \geq 1/2$, the last step because α^2 attains both 0 and 1. Conversely $\mathcal{A} \equiv 1/2$ achieves $\sup_{\alpha} |1/2 - \alpha^2| = 1/2$. The second equality is the same argument applied to constant predictors. $\square$

Where the content is. The argument is three lines because Theorem 2 has already done the work: once two training targets induce the *same* edge law, no function of that law can separate them. The construction is the claim; the proof is bookkeeping, and we do not present it as more than that.

Three remarks. (a) The bound is free of n : it is an information deficiency, not an estimation problem, and $n = \infty$ does not help. (b) It uses only that the *distribution* of $\mathcal{A}$ is α -free, so it also covers statistics given the population law or the conditioning c . (c) The value $1/2$ is attained, so the best data-only predictor is the one that ignores the data; it is not a bound we failed to sharpen.

What the theorem does not say. It does not say ρ is unknowable. ρ uses the coupling, and the coupling is available: every training step of a flow model draws a paired batch $(x_0^{(i)}, x_1^{(i)})$. Estimating ρ from those pairs costs nothing extra and can be done before the one-step model exists. The theorem says the pairing is not optional—it is the entire content.

4 WHAT DOES DECIDE: TWO CHANNELS

4.1 CHANNEL 1: CHANGE THE COUPLING

Corollaries 1–2 say the coupling alone moves ρ^* from 0 to 1. This is the mechanism behind every published remedy we know of: mini-batch OT coupling (Tong et al., 2023), reflow (Liu et al., 2022), consistency distillation (Song et al., 2023), and self-distilled one-step flow policies (Li et al., 2026a) all replace an independent coupling with an approximately deterministic one. Table 1 measures six method families plus two one-step-only variants under one protocol—same data, same architecture, same budget—and shows the split is predicted by the coupling and nothing else.

4.2 CHANNEL 2: THE LOSS FORM, AND WHAT DECIDES THE SPREAD

The less explored channel is to keep the independent coupling and change the loss. Let $q_c = f(\cdot, c)_{\#} \mathcal{N}(0, I_d)$ denote the law the one-step model produces for condition c , and let the training batch contain M paired samples $\{x_0^{(i)}, x_1^{(i)}, c^{(i)}\}_{i=1}^M$ with $G_i = f(x_0^{(i)}, c^{(i)})$.

Theorem 4 (Three-level ladder for the loss). *Under the independent coupling, as $M \rightarrow \infty$: (L1) the pointwise loss $\frac{1}{M} \sum_i \|G_i - x_1^{(i)}\|^2$ has unique population minimiser $q_c = \delta_{m(c)}$, giving $\rho = 0$; (L2) the unbalanced collection loss, with both arg mins taken within the condition,*

$$\frac{1}{M} \sum_i \min_j \|x_1^{(i)} - G_j\|^2 + \frac{1}{M} \sum_j \min_i \|G_j - x_1^{(i)}\|^2$$

converges to $\mathbb{E}_{y \sim p_c} [\min_{x \in \text{supp } q_c} \|y - x\|^2] + \mathbb{E}_{x \sim q_c} [\min_{y \in \text{supp } p_c} \|x - y\|^2]$, which vanishes iff $\text{supp } q_c = \text{supp } p_c$, and is otherwise silent about weights; (L3) the balanced collection loss, $\min_{\sigma} \frac{1}{M} \sum_i \|G_i - x_1^{(\sigma(i))}\|^2$ over bijections σ within each condition, converges to $W_2^2(p_c, q_c)$, whose unique minimiser is $q_c = p_c$ and $\rho = 1$.

Proof sketch. L1 is Cor. 1. For L2, the two terms are non-negative and vanish exactly when each support is contained in the other’s closure; no term depends on the mass assigned to a region once it is non-empty. For L3, minimising over bijections is exactly the optimal-transport problem between the empirical measures, whose $M \rightarrow \infty$ limit is W_2^2 , minimised uniquely at $q_c = p_c$. Appendix A. $\square$

The L2/L3 distinction is about the *set* of optima: L2 leaves the conditional weights free, L3 pins them—a difference the IMLE/Chamfer literature (Li & Malik, 2018) does not, to our knowledge, separate. But a free parameter is not a used one: whether training *exploits* that freedom is empirical, and Sec. 5.4 finds it does not—what flattens the conditional spread is the *scope* of the matching, not the absence of a bijection.

5 EXPERIMENTS

Common design. All comparisons hold data, architecture (MLP, 256×4 , input $[x, t, \text{onehot}(c)]$), optimiser, batch size and step count fixed and vary exactly one factor. Metrics are the spread ratio ρ of (1) (with a k NN estimator, Sec. 5.5), mode coverage (a mode counts as covered if any sample falls within 0.6 of its centre), and conditional sliced W_2 (cSW) normalised by $\sqrt{\text{tr Var}(x_1 | c)}$. Every cell uses ≥ 5 seeds. All runs are CPU-only.

5.1 $\rho^*(\alpha) = \alpha^2$ WITH THE DATA HELD FIXED

We instantiate (3) with 4 conditions, each an 8-mode ring of width σ , and T the 512-step Euler flow map of the exact conditional velocity field. We sweep $\alpha \in \{0, 0.25, 0.5, 0.75, 1\}$ with a *fresh* independent sample at every level: had we instead reused one realisation, the data-side statistics would be constant by construction and the test would restate the theorem instead of testing it. The k NN estimate of ρ moves $0.05006 \rightarrow 0.1054 \rightarrow 0.2854 \rightarrow 0.5747 \rightarrow 0.9766$, against the analytic α^2 and its finite- k correction $\alpha^2 + (1 - \alpha)^2/k$; the mean absolute deviation from the corrected prediction is 0.01405 (criterion < 0.08). Over the same sweep, the data-side statistics move by at most 0.002993 ($\hat{D}$) and 0.00028 ($\hat{\Lambda}$)—i.e. by Monte-Carlo noise—while ρ^* moves by 1.0 (Fig. 1a). All five pre-registered criteria pass (verdict PASS).

5.2 A BATTERY OF DATA-SIDE STATISTICS IS FLAT IN α

Theorem 3 is exact and needs no experiment; what needs checking is its empirical premise—that statistics one would actually reach for are, in fact, flat. We test seven: $\hat{D}$ (unexplained conditional-variance fraction), $\hat{\Lambda}$ (Gaussian deficit), the number of clusters selected by Calinski–Harabasz, mean nearest-neighbour distance, the k -means between/within variance ratio at $K = 8$, excess kurtosis, and the marginal trace variance.

Because these statistics differ by two orders of magnitude in their own sampling noise, a fixed relative threshold tests the noise rather than the α effect: run with naïve k -means initialisation, the $K = 8$ ratio varies from 15 to 36 *across seeds at the same α* . We therefore test with a one-way ANOVA over α with seeds as replicates (and with restarted k -means++). At level 0.05, no statistic shows a detectable α effect; the largest relative deviation from $\alpha = 0$ is 0.06374. By contrast the paired estimator’s mean absolute deviation from α^2 is 0.01405— $35.58\times$ smaller than the 0.5 that Theorem 3 imposes on *any* data-only predictor.

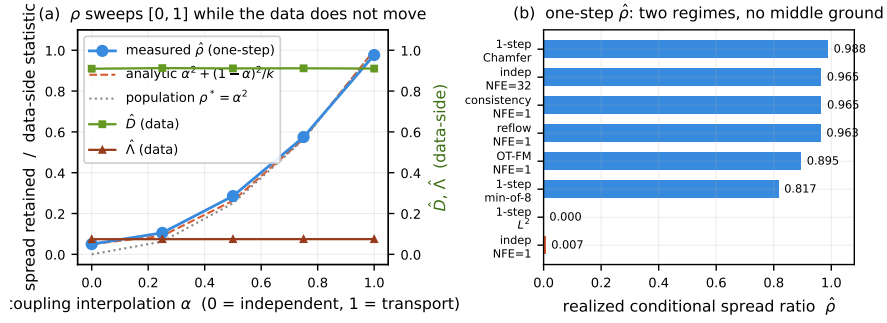


Figure 1: **The data cannot decide.** (a) ρ against α on the family of Thm. 2: measured (circles) versus the analytic α^2 (solid) and its finite- k correction (dashed). On the right axis, two data-side statistics over the same sweep: they are flat while ρ traverses $[0, 1]$. (b) Method families at one step, same data and budget; the training target—not the sampler—separates them.

Table 1: Method families at one step. Same data, architecture, optimiser, batch size and step count; only the training target differs. Reading the table against (2): *every* collapsing row has both an independent coupling and a pointwise loss, and removing either conjunct removes the collapse. “loss” distinguishes a pointwise L^2 objective from a collection-level one.

method (all at NFE= 1)	coupling	loss	ρ	modes / 8
flow matching	independent	pointwise	0.00669 ± 0.00159	0.05
same field, NFE= 32	independent	pointwise	0.9646 ± 0.009531	8
+ mini-batch OT	$\approx \text{det.}$	pointwise	0.8949 ± 0.0164	8
+ reflow	deterministic	pointwise	0.9631 ± 0.008115	8
distillation of the 32-step map	deterministic	pointwise	0.9654 ± 0.01008	8
one-step regression	independent	pointwise	$9.815 \times 10^{-5} \pm 8.811 \times 10^{-5}$	0
one-step, min-of- M	independent	collection	0.8165 ± 0.02117	8
one-step, bidirectional Chamfer (pooled)	independent	collection	0.9882 ± 0.07542	7.6

Sample size does not rescue the data side. Increasing n from 1000 to 16000 leaves the $\alpha = 0$ -versus- $\alpha = 1$ difference of every $[0, 1]$ -scaled statistic below 0.003—that is, at most 0.3% of the 1.0 that a useful predictor must resolve—while the paired estimator, with the same budget of samples, tracks α^2 . If a data-side statistic carried a real α -signal, its difference would converge to a nonzero constant as n grows; it does not. (Two of the seven statistics, the Calinski–Harabasz count and the k -means variance ratio, are ratio-scaled—their typical values are 8 and 35—so an absolute threshold is not meaningful for them; their α -independence rests on the scale-free ANOVA above.)

5.3 WHAT DOES DECIDE: THE METHOD MAP

Table 1 runs six method families plus two one-step-only variants on the same conditional ring mixture, and the outcome tracks (2) rather than the coupling alone. Every row that collapses has *both* an independent coupling and a pointwise loss: flow matching at $N = 1$ gives $\rho = 0.00669 \pm 0.00159$ with coverage 0.05/8, and a direct one-step regression gives 9.815×10^{-5} . Removing *either* conjunct removes the collapse—by the coupling (mini-batch OT 0.8949, reflow 0.9631, distillation 0.9654) or by the loss form with the coupling left independent (pooled bidirectional Chamfer 0.9882; re-measured as L2a in Table 2 at 0.9474, where its support/weight failure is dissected). The same field that collapses at $N = 1$ recovers at $N = 32$ (0.9646), so this is not capacity or optimisation.

5.4 THE LOSS LADDER: THE SCOPE OF THE MATCHING DECIDES

We train four one-step models on the *same* independent coupling, changing only the loss: pointwise L^2 (L1); a bidirectional Chamfer with both arg mins over the *whole batch*, as such losses are usually written (L2a); the same objective confined to each condition (L2b); and a balanced within-condition assignment (L3). L2b and L3 differ in exactly one respect—whether the assignment is a bijection—

Table 2: Loss ladder at one step, independent coupling held fixed. L1 collapses. L2a (Chamfer pooled over the batch) escapes and is indistinguishable from L2b on the aggregate ρ , yet flattens the conditional spread; L2b, matched within each condition, restores it; L3, balanced within each condition, is closest on conditional sliced W_2 . $\text{disp} = \text{std}_j(\rho_j)$; $|\rho_j - 1|$ is the mean absolute per-condition spread error. L2b and L3 differ only in whether the assignment is a bijection.

loss	ρ	modes / 8	disp	$ \rho_j - 1 $	cSW
L1 pointwise L^2	$9.716 \times 10^{-5} \pm 8.884 \times 10^{-5}$	0	0.0001205	0.9999	0.7024
L2a unbalanced, pooled over cond.	0.9474 ± 0.04917	7.8	0.5082	0.4331	0.3973
L2b unbalanced, within cond.	0.972 ± 0.01528	8	0.01885	0.0271	0.1294
L3 balanced, within cond.	0.8991 ± 0.02395	8	0.01805	0.09696	0.07917

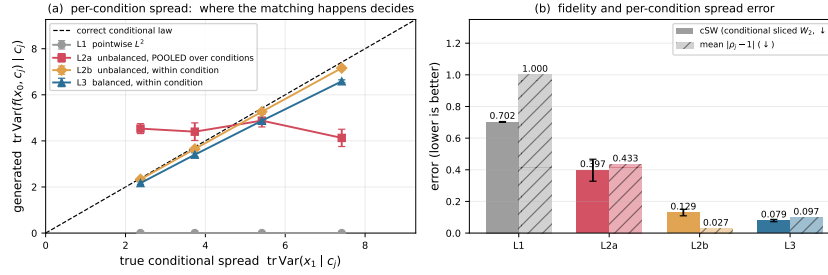


Figure 2: **The loss ladder (Thm. 4), observed.** Independent coupling held fixed; only the loss form changes. (a) Generated versus true per-condition spread, over four conditions whose true spread spans a factor of 3.1; the diagonal is correct. L1 (pointwise L^2) sits at zero; L2a (pooled Chamfer, whole-batch arg min) is nearly *horizontal*—support recovered, spread flattened; L2b (matched *within* each condition) returns to the diagonal; L3 (balanced, within condition) is closest in sliced W_2 . So the flattening comes from the scope of the matching, not from the missing bijection. (b) Conditional sliced W_2 and mean per-condition spread error.

the single variable Thm. 4 isolates; L2a versus L2b isolates a second, the *scope*. The conditions are four rings of radius 1.5, 1.9, 2.3, 2.7, so $\text{trVar}(x_1 | c)$ ranges by a factor of 3.1, making “support correct, weights wrong” directly visible.

Table 2 does *not* confirm the ladder as stated. Fig. 2a plots generated against true conditional spread. L1 sits at zero (collapse). L2a escapes, and on the aggregate it is indistinguishable from the method that gets the spread right: $\rho = 0.9474$ for L2a against 0.972 for L2b. The aggregate is simply blind here—what exposes L2a is that its spread is nearly *horizontal* (coefficient of variation 0.12 against a true 0.398), i.e. the dispersion of ρ_j across conditions: 0.5082 for L2a against 0.01885 for L2b. That horizontal reading does not survive the control: L2b, differing from L2a only in matching within each condition, returns to the diagonal, so the flattening came from the *pooling*, not from the missing bijection. L2b and L3 both track the diagonal, separated only by what the metric rewards: L3 has the lowest conditional sliced W_2 (0.07917 against 0.1294, a factor of 1.635) while L2b is nominally closer on the scale ratio ($|\rho_j - 1|$ 0.0271 against 0.09696). A single aggregate number cannot substitute for the conditional statistic: it reports 0.9474 for the pooled variant whose scale error is 0.4331, and 0.972 for the within-condition variant whose scale error is 0.0271.

A prediction that failed. We expected that dropping balancedness at fixed scope (L2b vs. L3) would flatten the conditional spread; it did not, and the ladder’s pre-registered verdict is **PARTIAL**—the check “balancedness reduces weight error” fails outright. The variable that did the flattening is the one we had not set out to vary, the scope of the arg min, so we state the theorem as a statement about scope first.

A corrected mechanism: min-of- M . min-of- M (IMLE) also escapes ($\rho = 0.8165$, coverage 8/8, against 9.815×10^{-5} for pointwise L^2) once its M candidates come from *independent* source noise. An earlier claim that it did not escape was an implementation artefact—repeating one source draw makes the objective literally pointwise L^2 (Appendix B(iii)).

5.5 ESTIMATING ρ FROM PAIRED SAMPLES

ρ^* is the variance of a conditional mean, so the natural estimator is k NN regression of x_1 on x_0 within each condition. This estimator has an analytic bias that we use as a check rather than a nuisance: under an independent coupling, the k -neighbour average of i.i.d. targets has variance $\text{Var}(x_1 | c)/k$, so $\hat{\rho}_{k\text{NN}}(k) \approx \rho^* + (1 - \rho^*)/k$, which at $\rho^* = 0$ is the $1/k$ law. Measured $k \cdot \hat{\rho}$ under the independent coupling is 1.002, 0.9775, 0.9369, 0.8885, 0.8257 at $k = 1, 2, 5, 10, 20$ —the law holds before the estimator is used for anything.

6 A NEGATIVE RESULT ABOUT OUR OWN EARLIER PROPOSAL

An earlier version of this line of work advanced $\hat{\Lambda}$, a null-calibrated conditional mode-separation score, as a *modality detector*. We report the experiment that kills that claim. On a fixed battery of ten population distributions the Gaussian deficit (the quantity behind $\hat{\Lambda}$) takes values 2.215×10^{-5} (Gaussian), 0.02529 (uniform ball), 0.02414 (Laplace), 0.0345 (heteroscedastic), 0.09003 (eight-mode ring), 0.1193 (Student- t_3), 0.1228 (log-normal), 0.1254 (exponential), 0.2257 (banana), 0.308 (two modes). It has the correct analytic ceiling of $1 - 2\sqrt{2/\pi} + 1$, which the finite-sample estimator already matches to 4.41×10^{-5} at $n = 5000$, and it is stable across sample sizes (largest disagreement between $n = 2000$ and $n = 2 \times 10^5$ is 0.008909). But it *sorts a single-mode banana above an eight-mode ring*: $0.2257 > 0.09003$. A statistic that ranks one mode above eight cannot be called a modality detector, whatever its construct validity on the remaining nine distributions (rank correlation 0.891 against a brute-force optimal-transport computation, and 0.905 against a constructed error measure).

We keep $\hat{\Lambda}$ as a descriptive index of departure from Gaussianity and drop the modality claim. This is not incidental to the paper: it is the same lesson as Sec. 3 in miniature. Any scalar summary of a conditional law—ours included—is a projection, and the projection that separates the cases you had in mind is not the projection that separates the cases that occur.

7 RELATED WORK

Few-step generative models. Consistency models (Song et al., 2023), consistency flow matching (Yang et al., 2024), shortcut models (Frans et al., 2024), mean flows (Geng et al., 2025), progressive (Salimans & Ho, 2022) and distribution-matching (Yin et al., 2023) distillation, and InstaFlow (Liu et al., 2023b) all target few-step sampling. Our contribution is not another sampler but a statement about which design choice governs the outcome, and a proof that the data cannot substitute for that choice.

Coupling and straightness. Rectified flow (Liu et al., 2022) shows straightening the coupling makes one step viable; stable diffusion 3 (Esser et al., 2024) shows the training-time distribution matters as much as the sampler; non-Gaussian sources reduce path crossing in physical simulation (Zhou & Jelovica, 2026). The coupling literature (Tong et al., 2023; Liu et al., 2022; Pooladian et al., 2023) already knows the coupling matters, and we do not claim otherwise—Thm. 1 is the tower property. We add (i) that it is not recoverable from data, so “the coupling matters” cannot become a data-side diagnostic (Thm. 3); and (ii) that the loss form is a second lever, governed by the *scope* over which candidates compete (Thm. 4).

One-step policies in RL. (Nguyen et al., 2026; Li et al., 2026a; Zhan et al., 2026; Im et al., 2026; Li et al., 2026b; Lee et al., 2026) report strong one-step policies, but (Li et al., 2026a; Chen et al., 2026) find the advantage disappears when the observation is weakened or the action chunk lengthened. (Chen et al., 2026) calls this a condition–target *structure* property and formalises it as an irreducible velocity loss. Sharper: that loss is a property of the training target, and the data enters only through the denominator of (1).

Collection-level losses. Implicit maximum likelihood estimation (Li & Malik, 2018) and Chamfer-style objectives are standard tools for avoiding mode-averaging, typically presented as

interchangeable. Thm. 4 separates them by the *scope* of the matching: only a balanced, within-condition assignment is pinned to the full W_2 objective, and Sec. 5.4 shows the scope—not the balance—is what flattens the conditional spread.

Diagnostics. (Li & Boström, 2026) gives a pre-training, conditional-spread-based few-step diagnostic (q^*/σ^*) for conformal prediction; $\sqrt{\hat{D}}$ is our version of that x -axis with a different normalisation, and we say so. We prove the diagnostic cannot be extended to the general one-step-adequacy question.

8 CONCLUSION, LIMITATIONS, AND WHAT WE WOULD WANT FALSIFIED

Whether one step is enough is decided by the training target—the coupling and the loss—not by the data. We proved no statistic of the data predicts it, with an exact minimax value of $1/2$; showed ρ is computable for free from the paired samples a trainer already draws; and mapped the two channels, including a loss channel set by the *scope* of the matching (pooled versus per-condition) and then by its balance.

Limitations. (1) The impossibility is proved for an explicit family—not contrived ($\alpha = 0, 1$ are the couplings in use) but not literally a practitioner’s target—and bounds the worst case. (2) Measurements are CPU-scale controlled *synthetic* families; we have not validated on D4RL (Fu et al., 2020), OGBench (Park et al., 2024), LIBERO (Liu et al., 2023a), or a VLA. (3) Balanced assignment costs a Hungarian solve per batch per condition; its scaling is not studied. (4) ρ measures conditional spread, not task reward.

What would falsify us. A data-side statistic that provably separates two training targets sharing an edge law; a deterministic coupling that nonetheless collapses at one step under pointwise L^2 (contradicting Cor. 2); or a demonstration that unbalanced and balanced collection losses converge to the same conditional law as $M \rightarrow \infty$ (contradicting Thm. 4).

AI USE STATEMENT

Generative AI systems were used to propose and refine the hypothesis, locate and summarise literature, critique mathematical claims, implement and audit code, suggest experiments, and draft text and figures. The authors are responsible for the complete submission; every AI-assisted claim and artefact was verified against the released code and result files, and the resulting corrections are recorded in Appendix B and the audit log.

REPRODUCIBILITY STATEMENT

Appendices C and D give the data, compute and execution protocol. The supplement contains the source code, per-run result files, table/figure generators, and regression tests for the failure modes in Appendix B. No number is hand-copied: every one is a macro generated from the released JSON.

ETHICS STATEMENT

This is an analysis of when one-step generative policies suffice. It uses only synthetic data with exactly controlled ground truth, involves no human subjects, and releases no new dataset. Where our own earlier proposal failed we say so (Sec. 6).

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A PROOFS

A.1 THEOREM 1

The minimiser of $\mathbb{E}\|f(x_0, c) - x_1\|^2$ over measurable f is the conditional expectation, by the L^2 projection theorem applied conditionally on (x_0, c) . For the range, condition on c and apply the law of total variance to the scalar components:

$$\mathrm{Var}(x_1 \mid c) = \mathbb{E}[\mathrm{Var}(x_1 \mid x_0, c) \mid c] + \mathrm{Var}(\mathbb{E}[x_1 \mid x_0, c] \mid c) \succeq \mathrm{Var}(\mathbb{E}[x_1 \mid x_0, c] \mid c)$$

in the positive-semidefinite order, so the trace of the second term is at most that of the first and $\rho \leq 1$; $\rho \geq 0$ is immediate. Equality in the trace holds iff $\mathrm{Var}(x_1 \mid x_0, c) = 0$ a.s., i.e. iff the coupling is deterministic given c .

A.2 THEOREM 3, FULL STATEMENT

The proof in the main text is complete; we record two extensions. (i) *Randomisation does not help*: if $\mathcal{A}$ has access to auxiliary randomness $U \perp\!\!\!\perp$ data, the law of (data, U) is still α -free, so the same Jensen argument applies to $\bar{a} = \mathbb{E}[A]$. (ii) *Knowledge of $p(c, x_1)$ does not help*: a statistic of the population law is a constant, and constants are covered by the second equality in the theorem. (iii) The same argument gives $\inf_{\mathcal{A}} \sup_{\alpha} \mathbb{E}(\mathcal{A} - \rho^*)^2 = 1/4$ for squared loss, attained at $1/2$ as well.

A.3 THEOREM 4, L3 LIMIT

For a batch of M samples within one condition, minimising $\frac{1}{M} \sum_i \|G_i - x_1^{(\sigma(i))}\|^2$ over bijections σ is exactly the optimal-transport problem between the empirical measures $\hat{q}_c$ and $\hat{p}_c$ with quadratic cost; its value is $W_2^2(\hat{q}_c, \hat{p}_c)$. As $M \rightarrow \infty$ the empirical measures converge weakly and $W_2^2(\hat{q}_c, \hat{p}_c) \rightarrow W_2^2(q_c, p_c)$ under the usual moment conditions. The unique minimiser over probability measures is $q_c = p_c$; the minimiser over *pushforwards of $\mathcal{N}(0, I_d)$ by a measurable map* is any f with $f(\cdot, c)_{\#} \mathcal{N}(0, I_d) = p(\cdot | c)$, and every such f has $\rho = 1$.

B CORRECTIONS WE MADE DURING THE STUDY

We record the errors we found and fixed, because each is a trap a reader re-implementing this will also hit.

(i) Never use a guessed effect size as a pre-registered criterion. Our first version of the $\hat{\Lambda}$ validation required $\Lambda(\Delta = 6) \geq 0.50$; it failed, but the failure said nothing about the statistic—the analytic ceiling of the quantity is 0.404, so the criterion was unsatisfiable. Likewise the first version of the loss-ladder criterion required $\text{disp}(\text{Chamfer}) > 2 \times \text{disp}(\text{balanced})$ and failed at a 40-step smoke test before the models had converged. Effect sizes depend on how far optimisation has gone and are not knowable in advance; we now pre-register only *directional* and *qualitative* predictions and report magnitudes descriptively.

(ii) Test invariance against the statistic’s own noise. Our first invariance check flagged the k -means between/within ratio as α -dependent (relative deviation 0.16). It is not: with naïve random initialisation the statistic varies from 15 to 36 *across seeds at fixed α* . A fixed relative threshold tests noise, not signal. We switched to a one-way ANOVA with seeds as replicates and to restarted k -means++.

(iii) Two collection-level objectives could not test their own claim. (a) *min-of- M* . Our first implementation of the min-of- M row built its M candidates by repeating a single source draw, $x_0^{(i)}, \dots, x_0^{(i)}$. A one-step network is a deterministic map, so the M outputs coincide *at every training state*—not only at collapse; the arg min is therefore constant and the loss is the pointwise L^2 loss on the same batch. The two code paths were identical to the last bit: $\max |\Delta w| = 0$ after 200 steps of each, and in the full runs the spread ratio agreed to all 17 printed digits (0.00010386879583898434 under both). The original row therefore measured nothing about IMLE. The corrected objective draws M *independent* x_0 per target, which is the original IMLE semantics, and Table 1 uses the re-generated row. (b) *the unbalanced Chamfer row*. Our first implementation took both arg mins over the *whole batch*, whereas the balanced row took its assignment *within each condition*. The comparison therefore varied two things at once—balancedness and conditioning scope—while Thm. 4(L2–L3) isolates the first. The corrected row matches within each condition (pooling is a *weaker* constraint, since a minimum over a superset is smaller, so this change strengthens the support requirement rather than manufacturing the support recovery we observe). The general lesson for both: an objective whose entire purpose is to *choose* among candidates should assert in code that the candidate sets and the choice are the ones the theorem talks about.

(iv) $\hat{\Lambda}$ is not a modality detector. Sec. 6. It sorts a single-mode banana (0.2257) above an eight-mode ring (0.09003).

(v) **A tolerance half the mode spacing silently inverts the mode metric.** Defining “on a mode” as within half the nearest-centre distance shrinks the tolerance below the mode width as K grows; at $K = 1$ it is undefined and hides the collapse. We use a σ -scale ball measured from mode *width*.

(vi) **Marginal sliced W_2 conflates the one-step floor with mode collapse.** The ratio SW@1/SW@32 reports a penalty even for an under-fit field that has none to report. The conditional ratio cSW@1/cSW@32 is the one the floor bounds, and we use it throughout.

(vii) **The no-free-parameter prediction used the W_1 constant inside a W_2 argument.** An earlier draft used $\sigma\sqrt{2/\pi}$ (the W_1 distance, equivalently the mean absolute deviation) where the W_2 distance is the scale gap σ , giving $\kappa_d = \sqrt{2/(\pi d)}$ instead of $1/\sqrt{d}$.

(viii) **“Coverage exactly zero” is not a satisfiable criterion.** The method map pre-registered “the independent coupling covers 0 of 8 modes at $N = 1$ ”. The theory (Cor. 1) is about the *exact* optimum, which does sit on $m(c)$ and covers exactly 0; a *trained* network cannot attain the exact optimum, and over 2000 evaluation samples an occasional point falls inside a mode’s tolerance ball, giving a mean coverage of 0.08/8. The criterion as written asked for numerical perfection. We relaxed it to at most 1 of 8 modes—still two orders of magnitude away from the passing side (8/8)—and record the change rather than quietly keeping the original verdict.

C RUNTIME AND REPRODUCIBILITY

Every number in the main text is produced by the scripts under `code/` from the raw per-run JSON files under `results/` and `logs/`; the paper never contains a hand-copied statistic. Numbers in the theory sections come from `paper/numbers.theory.tex`, generated by `code/analysis/make_theory_macros.py`, whose every macro carries a comment naming the JSON field it came from.

D ESTIMATORS, AND WHY THEY ARE DELIBERATELY CLASSICAL

Every diagnostic in this paper is built from standard, assumption-light non-parametric tools, so that a reader can re-implement it without training any auxiliary model. The conditional mean behind ρ is estimated by k NN regression (Cover & Hart, 1967); the mode count and the between/within variance ratio use k -means (Lloyd, 1982; MacQueen, 1967) selected by the Calinski–Harabasz criterion (Caliński & Harabasz, 1974); the conditional spread is measured by sliced W_2 (Kolouri et al., 2019). We use these classical estimators on purpose: they carry almost no assumptions, need no hyper-parameter search, and are *decoupled from the generative model being diagnosed*, which is what lets ρ be defined and computed before any one-step model exists (Sec. 5.5). The one consequence to keep in mind is the k NN bias in (1): all statements are about the population quantity ρ^* , and we report the estimator’s analytic bias rather than correcting it away.

E SCOPE AND ASSUMPTIONS

- **Field:** machine learning — generative modelling for decision making, as an *analysis and diagnosis* contribution, not a new sampler.
- **Tasks:** synthetic conditional ring mixtures with closed-form ground truth; a multi-goal continuous-control task used as an implementation smoke test only. Conditional MNIST 16×16 (LeCun et al., 1998) was trained in an earlier iteration of this work under a superseded protocol and is *not* reported here: at our CPU budget the field was under-fitted, so those runs could neither confirm nor refute anything. No number in this paper depends on them.
- **Datasets:** entirely synthetic; no proprietary data, no robot hardware, no pretrained backbone.
- **Compute:** CPU only (a 20-core workstation, PyTorch `cpu` build). No GPU was used at any point.

- **Attribution:** we do not reuse any author repository; each component is re-implemented from the published objective. Any error is ours.
- **Target venue:** ICLR 2027, double-blind, main text ≤ 9 pages excluding references and appendices (Committee, 2026).